

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=milk

retailer=silpo

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	0.012483062457844212	0.01	4.729732285868142e-13	0.1	1.068911264269473e-21	0.1	I(1)	1.0	milk	silpo	promo_controlled	nan	nan
prozorro	0.0014111002754477221	0.01	3.8189780115708437e-13	0.1	4.4125656034640205e-11	0.1	I(1)	0.0	milk	silpo	promo_controlled	nan	nan
retail	1.8563080372772102e-09	0.02253830533462102	1.862588539502213e-08	0.1	0.0007337482414083765	0.1	I(1)	0.0	milk	silpo	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo	promo_controlled	0.07470887303151186	201.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo	promo_controlled	3.9974963442764655e-05	37.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo	promo_controlled	0.9816231810873596	48.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
milk	silpo	promo_controlled	daily	producer_to_prozorro	ARDL	prozorro	producer	201	0.6442401315402435	-0.28431662591594176	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=0.001676; ...	Primary chain is strictly...
milk	silpo	promo_controlled	daily	producer_to_prozorro	ECM	prozorro	producer	197	-0.0021550039692882494	-0.3006834081052232	-0.9538372381620321	3.2952288002158942e-21	nan	nan	nan	unreliable	1	engle_granger_p=1.912e-24...	Primary chain is strictly...
milk	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	199	0.014643405379010765	-0.7451033547254802	-1.0911455790247266	2.2107645052270345e-26	0.9865928543792826	0.016656976279982996	nan	unreliable	1	Wald short/long asymmetry...	Primary chain is strictly...
milk	silpo	promo_controlled	daily	prozorro_to_retail	ECM	retail	prozorro	35	0.037804208603540196	0.07959450904544814	-0.9002155090608192	0.0004638981008127678	nan	nan	nan	ok	0	engle_granger_p=9.326e-06...	Primary chain is strictly...
milk	silpo	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	35	0.13095882303016856	0.11453901982630346	-1.1487124706777685	2.159825703741887e-05	0.16485894810185805	0.32238702802459634	nan	ok	0	Wald short/long asymmetry...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
ARDL	producer_to_prozorro	prozorro	producer	0.997947032879507	0.20798453556018956	2.474090685540177e-48	0
ECM	producer_to_prozorro	prozorro	producer	0.15369243238216118	0.0021398456723945958	4.744864281639144e-221	1
NARDL	producer_to_prozorro	prozorro	producer	0.14771696529720213	0.0004696321631414275	8.675957601562549e-203	1
ECM	prozorro_to_retail	retail	prozorro	0.9838835608618121	0.7600112143069808	0.04648178311126383	0
NARDL	prozorro_to_retail	retail	prozorro	0.9821137350650245	0.7475783559300255	0.11121730903409086	0

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2021-02-01	4.889813923695362	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-02	4.880939529121857	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-03	4.873865236176802	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-04	4.868432562653949	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-05	4.864483026347049	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-06	4.861858145049856	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-07	4.860399436556123	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-08	4.8599484186596005	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-09	4.861523687586162	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-10	4.865852176359705	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-11	4.872337907971017	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-12	4.880384905410883	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-13	4.889397191670094	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-14	4.898778789739431	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-15	4.907933722609686	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-16	4.919261740789084	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-17	4.93424761149779	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-18	4.950922882794503	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-19	4.967319102737919	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-20	4.981467819386735	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-21	4.991400580799647	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-22	4.995148935035351	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-23	4.937254183107472	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-24	4.7879466649776745	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-25	4.583791487126728	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-26	4.361353756035397	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-27	4.157198578184451	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-28	4.0078910600546545	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-01	3.949996308126774	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-02	3.9561770869121475	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-03	3.972116990095481	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-04	3.9939123679175887	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-05	4.017659570619289	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-06	4.039454948441397	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-07	4.05539485162473	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-08	4.061575630410104	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-09	4.058274307819267	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-10	4.049435990260466	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-11	4.036659153054266	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-12	4.021542271521231	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1	-0.1619954987840599	
producer_to_prozorro	2	-0.1596960610741088	
producer_to_prozorro	3	-0.1550755940895887	
producer_to_prozorro	4	-0.14782606735994822	
producer_to_prozorro	5	-0.13755773452175912	
producer_to_prozorro	6	-0.12184188698467123	
producer_to_prozorro	7	-0.1052691917763927	
producer_to_prozorro	8	-0.08910616067044506	
producer_to_prozorro	9	-0.07590716773933302	
producer_to_prozorro	10	-0.06623846374994181	
producer_to_prozorro	11	-0.0604388966635031	
producer_to_prozorro	12	-0.053445772573508486	
producer_to_prozorro	13	-0.044442939499120704	
producer_to_prozorro	14	-0.03488503971137872	
producer_to_prozorro	15	-0.027271625330034636	
producer_to_prozorro	16	-0.02250000775439031	
producer_to_prozorro	17	-0.021139446186402542	
producer_to_prozorro	18	-0.017275889530894442	
producer_to_prozorro	19	-0.0052249478693689524	
producer_to_prozorro	20	0.012485029223047036	
producer_to_prozorro	21	0.03347787604499084	
producer_to_prozorro	22	0.05374998643553993	
producer_to_prozorro	23	0.06948873713684375	
producer_to_prozorro	24	0.07823993038535337	
producer_to_prozorro	25	0.08172443030355239	
producer_to_prozorro	26	0.08396988659951972	
producer_to_prozorro	27	0.0844840372389968	
producer_to_prozorro	28	0.08380704389180071	
producer_to_prozorro	29	0.08186704327105128	
producer_to_prozorro	30	0.07802717791370897	
prozorro_to_retail	1	0.018666903841372963	
prozorro_to_retail	2	0.02464182251638258	
prozorro_to_retail	3	-0.10689713085534079	
prozorro_to_retail	4	0.12835061477732507	
prozorro_to_retail	5	0.04117797867055295	
prozorro_to_retail	6	0.08740005404769866	
prozorro_to_retail	7	0.18619104902697076	
prozorro_to_retail	8	0.0505504637048315	
prozorro_to_retail	9	-0.02652991307560794	
prozorro_to_retail	10	-0.06240811812210943	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
milk	silpo	promo_controlled	daily	I(1)	I(1)	I(1)	1793	211	48	201	37	0	Admissible for ARDL/ECM s...

Table: NARDL_Multipliers

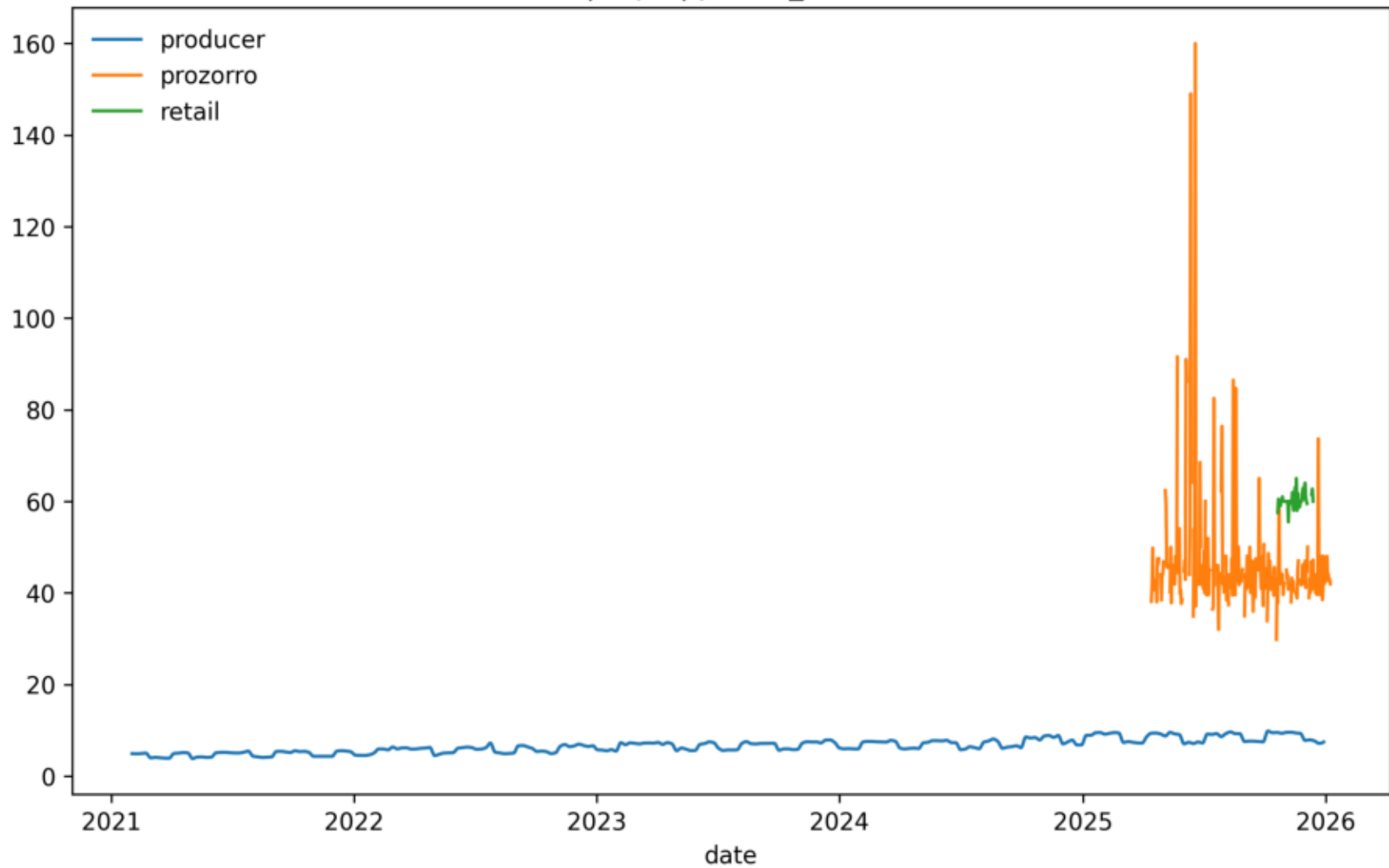
horizon	mult_pos	mult_neg	link
0	-1.7742938741039946	-1.7889372794830054	producer_to_prozorro
1	-1.8965525776203653	-1.912204994969057	producer_to_prozorro
2	-1.9049768813975803	-1.9206988252756965	producer_to_prozorro
3	-1.9055573627219329	-1.9212840973641223	producer_to_prozorro
4	-1.9055973611095542	-1.9213244258619995	producer_to_prozorro
5	-1.9056001172206387	-1.921327204719514	producer_to_prozorro
6	-1.9056003071320016	-1.9213273961982325	producer_to_prozorro
7	-1.9056003202179495	-1.9213274093921797	producer_to_prozorro
8	-1.905600321119644	-1.921327410301316	producer_to_prozorro
9	-1.9056003211817756	-1.9213274103639606	producer_to_prozorro
10	-1.9056003211860568	-1.9213274103682771	producer_to_prozorro
11	-1.905600321186352	-1.9213274103685745	producer_to_prozorro
12	-1.9056003211863721	-1.921327410368595	producer_to_prozorro
13	-1.9056003211863737	-1.9213274103685964	producer_to_prozorro
14	-1.9056003211863737	-1.9213274103685964	producer_to_prozorro
15	-1.9056003211863737	-1.9213274103685964	producer_to_prozorro
16	-1.9056003211863737	-1.9213274103685964	producer_to_prozorro
17	-1.9056003211863737	-1.9213274103685964	producer_to_prozorro
18	-1.9056003211863737	-1.9213274103685964	producer_to_prozorro
19	-1.9056003211863737	-1.9213274103685964	producer_to_prozorro
20	-1.9056003211863737	-1.9213274103685964	producer_to_prozorro
0	0.06345346241198797	-0.06750536061818059	prozorro_to_retail
1	0.06926473730126805	-0.07368772155075146	prozorro_to_retail
2	0.06979695279519194	-0.07425392231392884	prozorro_to_retail
3	0.06984569482467244	-0.0743057768251292	prozorro_to_retail
4	0.06985015877858773	-0.0743105258302643	prozorro_to_retail
5	0.0698505676020358	-0.07431096075963209	prozorro_to_retail
6	0.06985060504341957	-0.07431100059188107	prozorro_to_retail
7	0.06985060847242353	-0.07431100423984825	prozorro_to_retail
8	0.06985060878646288	-0.07431100457394098	prozorro_to_retail
9	0.06985060881522362	-0.07431100460453828	prozorro_to_retail
10	0.06985060881785762	-0.07431100460734048	prozorro_to_retail
11	0.06985060881809886	-0.07431100460759711	prozorro_to_retail
12	0.06985060881812095	-0.07431100460762062	prozorro_to_retail
13	0.06985060881812297	-0.07431100460762277	prozorro_to_retail
14	0.06985060881812316	-0.07431100460762297	prozorro_to_retail
15	0.06985060881812317	-0.07431100460762298	prozorro_to_retail
16	0.06985060881812317	-0.07431100460762298	prozorro_to_retail
17	0.06985060881812317	-0.07431100460762298	prozorro_to_retail
18	0.06985060881812317	-0.07431100460762298	prozorro_to_retail

Table: VECM_IRF

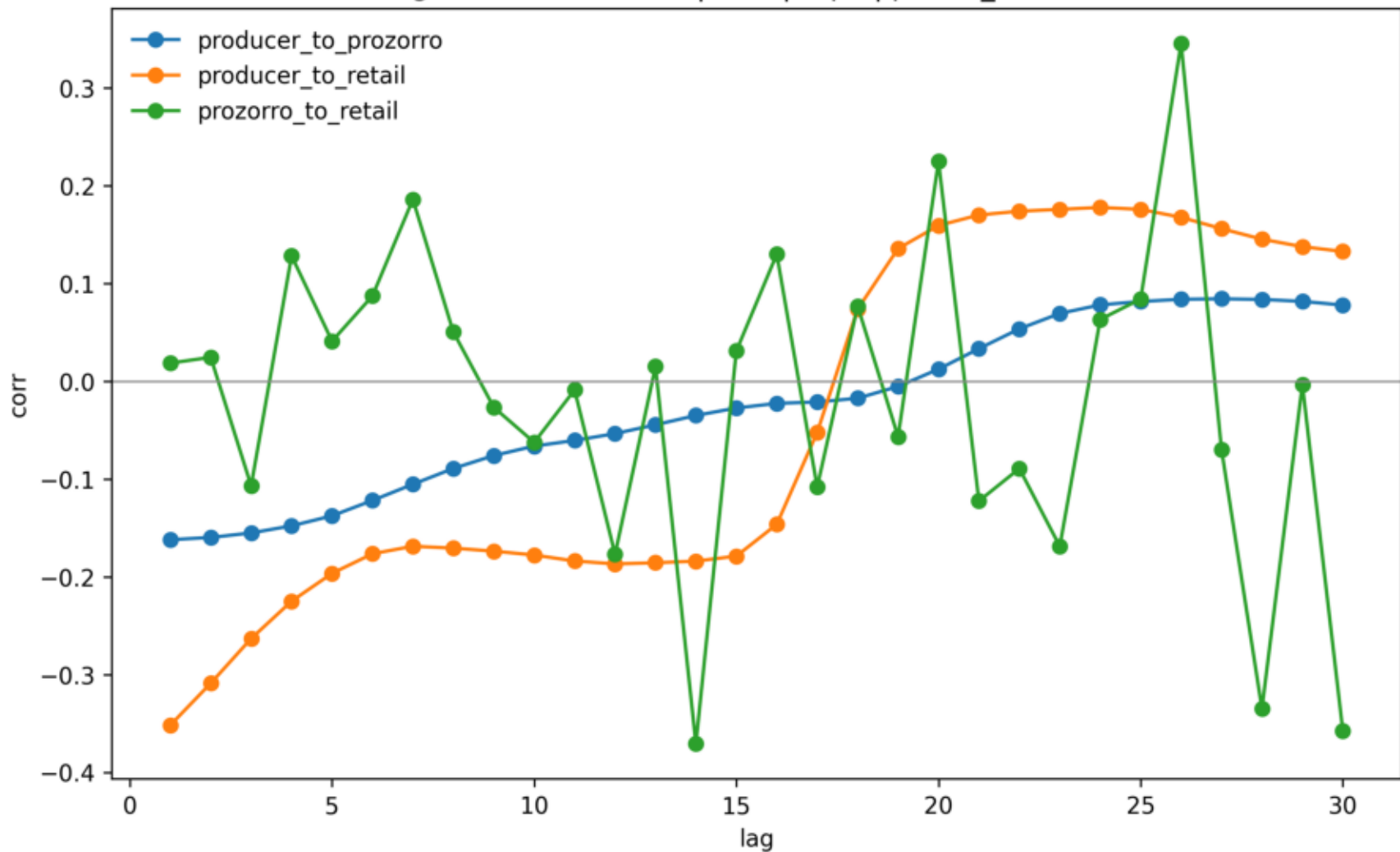
No rows

time_series_promo_controlled.png

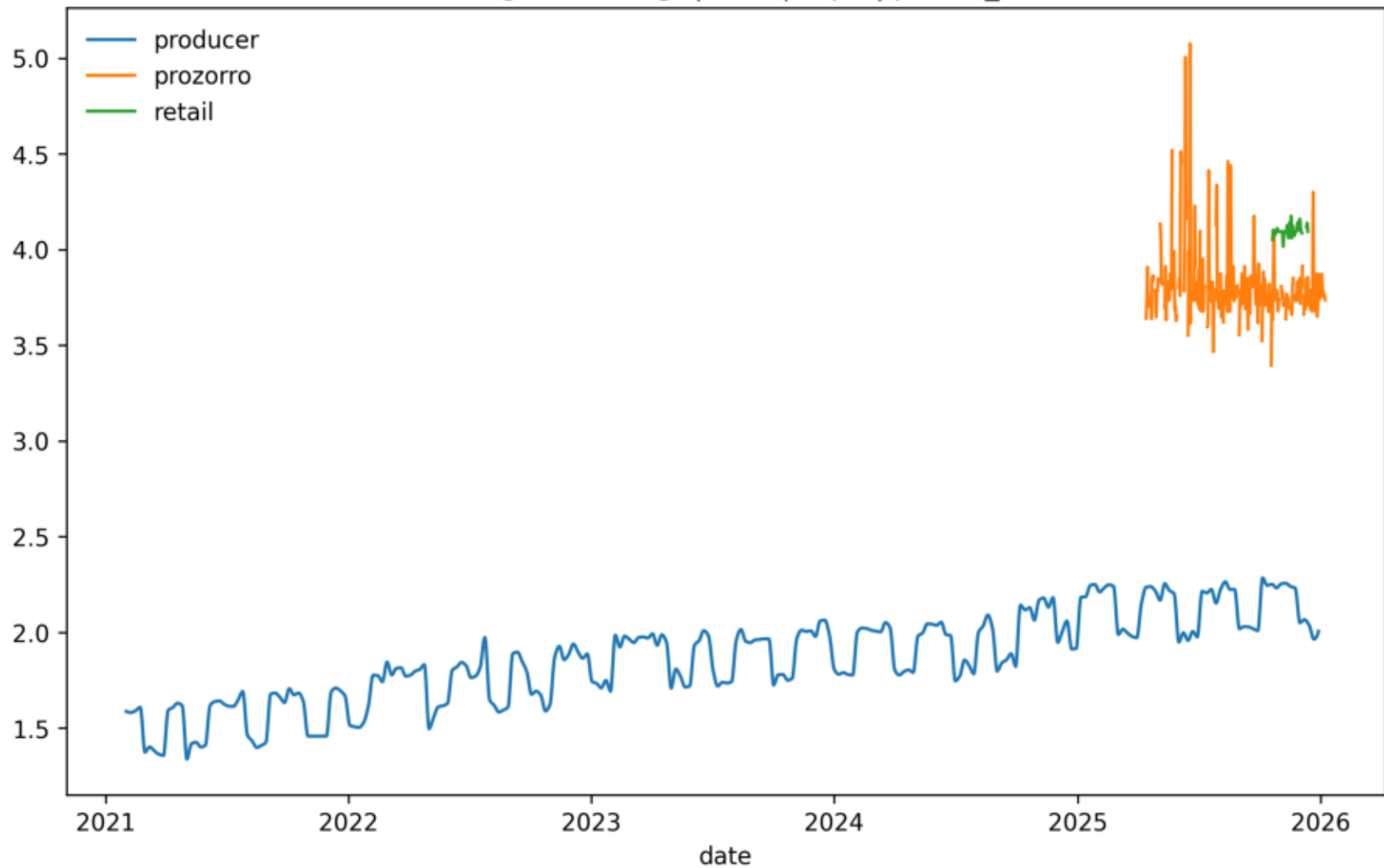
milk | silpo | promo_controlled



Lag-Correlation Profile | milk | silpo | promo_controlled



ECM Level Alignment (log) | milk | silpo | promo_controlled



NARDL Dynamic Multipliers | milk | silpo | promo_controlled

